Lecture-10 [28 August 2026]

Value Iteration and its Variants

- Input $\lambda \in [0,1)$, $\varepsilon > 0$, $\sigma > \left(\frac{1-\lambda}{\lambda}\right)\varepsilon$.
- Initialise $\underline{v}' = \{v'(s) : s \in \mathcal{S}\}$
- While $\sigma \geq \left(\frac{1-\lambda}{\lambda}\right)\varepsilon$:
  - $\underline{v} \leftarrow \underline{v}'$.
  - Set $\underline{v}' \leftarrow L\underline{v}$ $\quad v'(s) = \max_a \left[r(s,a) + \sum_{s' \in \mathcal{S}} P(s'|s,a)v(s')\right]$
  - $\sigma \leftarrow sp(\underline{v}' - \underline{v}) = sp(L\underline{v} - \underline{v})$.
- Terminate and return $\phi_\varepsilon : \mathcal{S} \to \mathcal{A}$ and $V_{\infty,\lambda,\varepsilon} : \mathcal{S} \to \mathbb{R}$ as:

$$\phi_\varepsilon(s) = \arg\max_{a \in \mathcal{A}} \left[r(s,a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s,a)\, v(s')\right], \quad s \in \mathcal{S}$$

$$V_{\infty,\lambda,\varepsilon} = \underline{v}' + \frac{\lambda}{1-\lambda} \min(\underline{v}' - \underline{v})\, \underline{1}$$
$$= L\underline{v} + \frac{\lambda}{1-\lambda} \min(L\underline{v} - \underline{v})\, \underline{1}.$$

Remarks on value iteration

① $\underbrace{\text{Span-based stopping criterion}}_{sp(\underline{v}' - \underline{v}) < \left(\frac{1-\lambda}{\lambda}\right)\varepsilon}$ generally results in fewer iterations than $\underbrace{\text{norm-based stopping}}_{\|\underline{v}' - \underline{v}\|_\infty < \left(\frac{1-\lambda}{\lambda}\right)\varepsilon}$

$r_{min} := \min_{s \in \mathcal{S}} \min_{a \in \mathcal{A}} r(s,a)$

$\underline{v}_0 = \frac{r_{min}}{1-\lambda} \underline{1}$

② The above variant of value iteration is called synchronous value iteration because at every iteration, the value of every state is updated. We will see an asynchronous variant of value iteration below.

Theorem 5.16 [Convergence guarantees for value iteration]

Fix $\lambda \in [0,1)$ and $\varepsilon > 0$. For any starting $\underline{v}_0 \in \mathbb{R}^{|\mathcal{S}|}$, the sequence of iterates $\underline{v}_1, \underline{v}_2, \ldots$ generated by value iteration satisfies:

① $\|\underline{v}_n - \underline{v}^*_{\infty,\lambda}\|_\infty \to 0$ as $n \to \infty$.

(2) $sp\left(\underline{v}_n - v^*_{\infty,\lambda}\right) \longrightarrow 0$ as $n \to \infty$.

(3) There exists a finite epoch $N \in \mathbb{N}$ such that

$$sp\left(\underline{v}_{n+1} - \underline{v}_n\right) < \left(\frac{1-\lambda}{\lambda}\right)\varepsilon \qquad \forall\ n \geq N.$$

Thus, the value iteration algorithm stops after a finite number of iterations.

(4) Upon termination :

(a) $\| v_{\infty,\lambda,\varepsilon} - v^*_{\infty,\lambda} \|_\infty < \varepsilon$.

(b) $\| v^{\phi_\varepsilon}_{\infty,\lambda} - v^*_{\infty,\lambda} \|_\infty < \varepsilon$.

This confirms that the policy $\phi_\varepsilon \in \Pi^{SD}$ returned by value iteration is ε-optimal.

<u>Proof of Theorem 5.16</u>

Because $\underline{v}_{n+1} = L\underline{v}_n \quad \forall\ n \geq 0$, part (1) simply follows from the Banach fixed point theorem, noting that L is a contraction with the unique fixed point $\underline{v}^* = v^*_{\infty,\lambda}$.

Part (2) follows from part (1) and the observation that

$$sp\left(\underline{v}_n - v^*_{\infty,\lambda}\right) \leq 2 \|\underline{v}_n - v^*_{\infty,\lambda}\|_\infty.$$

Part (3) follows by noting that

$$\begin{aligned} sp\left(\underline{v}_{n+1} - \underline{v}_n\right) &= sp\left(L\underline{v}_n - L\underline{v}_{n-1}\right) \\ &\leq \lambda\gamma^* \, sp\left(\underline{v}_n - \underline{v}_{n-1}\right) \\ &\ \ \vdots \\ &\leq \left(\lambda\gamma^*\right)^n sp\left(\underline{v}_1 - \underline{v}_0\right) \qquad \forall\ n \geq 0. \end{aligned}$$

Choose N to be the smallest integer satisfying

$$\left(\lambda\gamma^*\right)^N sp\left(\underline{v}_1 - \underline{v}_0\right) < \left(\frac{1-\lambda}{\lambda}\right)\varepsilon.$$

Then, clearly, for all $n \geq N$,

$$\begin{aligned} sp\left(\underline{v}_{n+1} - \underline{v}_n\right) &\leq \left(\lambda\gamma^*\right)^n sp\left(\underline{v}_1 - \underline{v}_0\right) \\ &\leq \left(\lambda\gamma^*\right)^N sp\left(\underline{v}_1 - \underline{v}_0\right) < \left(\frac{1-\lambda}{\lambda}\right)\varepsilon. \end{aligned}$$

Part (4) is simply a consequence of Theorem 5.13 (see lecture 09 notes).

Theorem 5.17 [Convergence rates for value iteration]

Let $\{\underline{v}_n\}_{n\geq 0}$ be the sequence of iterates generated by value iteration.

Then, $\{\underline{v}_n\}$ converges to $v^*_{\infty,\lambda}$ geometrically, i.e.;

$$\| \underline{v}_{n+1} - v^*_{\infty,\lambda} \|_\infty \leq \frac{\lambda^n}{1-\lambda} \| \underline{v}_1 - \underline{v}_0 \|_\infty .$$

Proof of Theorem 5.17

$$\begin{aligned} \| \underline{v}_{n+1} - v^*_{\infty,\lambda} \|_\infty &= \| L\underline{v}_n - L v^*_{\infty,\lambda} \|_\infty \\ &= \| L\underline{v}_n - L\underline{v}_{n+1} + L\underline{v}_{n+1} - L v^*_{\infty,\lambda} \|_\infty \\ &\leq \| L\underline{v}_{n+1} - L\underline{v}_n \|_\infty + \| L\underline{v}_{n+1} - L v^*_{\infty,\lambda} \|_\infty \\ &\leq \lambda^n \| \underline{v}_1 - \underline{v}_0 \|_\infty + \lambda \| \underline{v}_{n+1} - v^*_{\infty,\lambda} \|_\infty . \end{aligned}$$

Rearranging terms, we arrive at the desired result.

Gauss-Siedel Value Iteration

Let $\mathcal{S} = \{s_1, \dots, s_M\}$ denote an enumeration of states of the state space.

The Gauss-Siedel method sequentially updates the value of each state by using the updated values for all previous states.

- Input $\lambda \in [0,1)$, $\varepsilon > 0$, $\sigma > \left(\frac{1-\lambda}{\lambda}\right)\varepsilon$.
- Initialise $\underline{v}' = \{v'(s) : s \in \mathcal{S}\}$.
- While $\sigma \geq \varepsilon\left(\frac{1-\lambda}{\lambda}\right)$:
  - $\underline{v} \leftarrow \underline{v}'$
  - For $k = 1, \dots, M$, compute

$$v'(s_k) \leftarrow \max_a \Big[r(s_k, a) + \lambda \Big(\sum_{j<k} P(s_j | s_k, a) \, v'(s_j)$$

$$+ \sum_{j \geq k} P(s_j | s_k, a) \, v(s_j) \Big) \Big].$$

- $\sigma \leftarrow \| \underline{v}' - \underline{v} \|_\infty$

– Terminate and return $\underline{v}'$ and policy $\psi_\varepsilon : \mathcal{S} \to \mathcal{A}$ as

$$\forall k : \quad \psi_\varepsilon(s_k) = \arg\max_a \Big[r(s_k, a) + \lambda \Big(\sum_{j<k} P(s_j | s_k, a) \, v'(s_j)$$
$$+ \sum_{j \geq k} P(s_j | s_k, a) \, v(s_j) \Big) \Big].$$

Remarks :

1. This algorithm updates values asynchronously.
2. This algorithm uses norm-based stopping. One may also use span-based stopping, but theoretical guarantees are only known at present for norm-based stopping.
3. The policy ψ_ε has been constructed to simply replace max in the update rule with arg max, to make the proofs of theoretical guarantees simpler. One may also return the policy

$$\phi_\varepsilon(s_k) \in \arg\max_a \Big[r(s_k, a) + \sum_{j=1}^{M} P(s_j | s_k, a) \, v'(s_j) \Big].$$

Proposition 5.8

Let $G : \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ be defined as the mapping

$$\forall k : \quad G\underline{v}(s_k) = \max_{a \in \mathcal{A}} \Big[r(s_k, a) + \lambda \Big(\sum_{j<k} P(s_j | s_k, a) \, G\underline{v}(s_j)$$
$$+ \sum_{j \geq k} P(s_j | s_k, a) \, v(s_j) \Big) \Big].$$

Then, G is a contraction with modulus at most λ.

Proof of Proposition 5.8

Fix arbitrary $\underline{u}, \underline{v} \in \mathbb{R}^{|\mathcal{S}|}$. Notice that

$$G\underline{v}(s_1) = \max_{a \in \mathcal{A}} \Big[r(s_1, a) + \lambda \sum_{j=1}^{M} P(s_j | s_1, a) \, v(s_j) \Big] = L\underline{v}(s_1).$$

Therefore,

$$|G\underline{v}(s_1) - G\underline{u}(s_1)| = |L\underline{v}(s_1) - L\underline{u}(s_1)|$$
$$\leq \| L\underline{v} - L\underline{u} \|_\infty$$
$$\leq \lambda \| \underline{v} - \underline{u} \|_\infty .$$

Assume that $|G\underline{v}(s_j) - G\underline{u}(s_j)| \leq \lambda \| \underline{v} - \underline{u} \|_\infty \quad \forall j = 1, \dots, k-1.$

(a) Case 1 : $G\underline{v}(s_k) \geq G\underline{u}(s_k)$:

Let $a^*_{\underline{v}}$ be any action such that

$$a^*_{\underline{v}} \in \arg\max_{a \in A} \Big[r(s_k, a) + \lambda \Big(\sum_{j<k} P(s_j | s_k, a)\, G\underline{v}(s_j) + \sum_{j \geq k} P(s_j | s_k, a)\, v(s_j) \Big) \Big]$$

Then, we have

$$0 \leq G\underline{v}(s_k) - G\underline{u}(s_k)$$

$$= r(s_k, a^*_{\underline{v}}) + \lambda \Big(\sum_{j<k} P(s_j | s_k, a^*_{\underline{v}})\, G\underline{v}(s_j) + \sum_{j \geq k} P(s_j | s_k, a^*_{\underline{v}})\, v(s_j) \Big) - G\underline{u}(s_k)$$

$$\leq r(s_k, a^*_{\underline{v}}) + \lambda \Big(\sum_{j<k} P(s_j | s_k, a^*_{\underline{v}})\, G\underline{v}(s_j) + \sum_{j \geq k} P(s_j | s_k, a^*_{\underline{v}})\, v(s_j) \Big)$$
$$- \Big\{ r(s_k, a^*_{\underline{v}}) + \lambda \Big(\sum_{j<k} P(s_j | s_k, a^*_{\underline{v}})\, G\underline{u}(s_j) + \sum_{j \geq k} P(s_j | s_k, a^*_{\underline{v}})\, u(s_j) \Big) \Big\}$$

$$= \lambda \Big(\sum_{j<k} P(s_j | s_k, a^*_{\underline{v}}) \big(G\underline{v}(s_j) - G\underline{u}(s_j) \big) + \sum_{j \geq k} P(s_j | s_k, a^*_{\underline{v}}) \big(v(s_j) - u(s_j) \big) \Big)$$

$|G\underline{v}(s_j) - G\underline{u}(s_j)| \leq \lambda \| \underline{v} - \underline{u} \|_\infty \leq \| \underline{v} - \underline{u} \|_\infty$

by induction hypothesis

$$\leq \lambda \Big(\sum_{j<k} P(s_j | s_k, a^*_{\underline{v}}) \| \underline{v} - \underline{u} \|_\infty + \sum_{j \geq k} P(s_j | s_k, a^*_{\underline{v}}) \big(v(s_j) - u(s_j) \big) \Big)$$

$$\leq \lambda \| \underline{v} - \underline{u} \|_\infty .$$

(b) Case 2 : $G\underline{v}(s_k) < G\underline{u}(s_k)$

Repeating the above arguments with the roles of $\underline{u}$ and $\underline{v}$ interchanged, we get

$$0 \le G\underline{u}(s_k) - G\underline{v}(s_k) \le \lambda \|\underline{v}-\underline{u}\|_\infty \qquad \forall k \in \{1, \dots, M\}.$$

Combining the above cases, we get

$$\left| G\underline{v}(s_k) - G\underline{u}(s_k) \right| \le \lambda \|\underline{v}-\underline{u}\|_\infty \qquad \forall k \in \{1, \dots, M\}$$

$$\Rightarrow \quad \|G\underline{v} - G\underline{u}\|_\infty \le \lambda \|\underline{v}-\underline{u}\|_\infty .$$

<u>Theorem 5.18</u> [Convergence properties of Gauss-Siedel value iteration]

Fix $\lambda \in [0,1)$ and $\varepsilon > 0$. Let $\{\underline{v}_n\}_{n\ge 0}$ denote the sequence of iterates generated by the Gauss-Siedel value iteration algorithm. Then :

1. $\|\underline{v}_n - v^*_{\infty,\lambda}\|_\infty \to 0$ as $n \to \infty$.
2. $sp\left(\underline{v}_n - v^*_{\infty,\lambda}\right) \to 0$ as $n \to \infty$.
3. There exists a finite epoch $N \in \mathbb{N}$ such that
$$\|\underline{v}_{n+1} - \underline{v}_n\|_\infty < \left(\frac{1-\lambda}{\lambda}\right)\varepsilon \qquad \forall n \ge N.$$
Thus, the Gauss-Siedel value iteration algorithm terminates after finitely many iterations.
4. Upon termination :
   (a) The policy $\psi_\varepsilon \in \Pi^{SD}$ returned is 2ε-optimal, i.e.,
   $$\|v^{\psi_\varepsilon}_{\infty,\lambda} - v^*_{\infty,\lambda}\|_\infty < 2\varepsilon .$$
   (b) For all $n \ge N$, where N is as in part (3) above,
   $$\|\underline{v}_n - v^*_{\infty,\lambda}\|_\infty < \frac{\varepsilon}{\lambda}, \qquad sp\left(\underline{v}_n - v^*_{\infty,\lambda}\right) \le \frac{2\varepsilon}{\lambda}.$$

Proof of Theorem 5.18

(1) Noting that $\underline{v}_{n+1} = G\underline{v}_n$ $\forall n \geq 0$, and using the fact that G is a contraction, we get that there exists a unique fixed point $\underline{v}^* \in \mathbb{R}^{|\mathcal{S}|}$ for which $\underline{v}^* = G\underline{v}^*$ and $\underline{v}_n \to \underline{v}^*$. Then, for each $k \in \{1, \dots, M\}$,

$$\begin{aligned} v^*(s_k) &= G\underline{v}^*(s_k) \\ &= \max_{a \in \mathcal{A}} \left[r(s_k, a) + \lambda \left(\sum_{j<k} P(s_j | s_k, a)\, G\underline{v}^*(s_j) + \sum_{j \geq k} P(s_j | s_k, a)\, v^*(s_j) \right) \right] \\ &= \max_{a \in \mathcal{A}} \left[r(s_k, a) + \lambda \left(\sum_{j<k} P(s_j | s_k, a)\, v^*(s_j) + \sum_{j \geq k} P(s_j | s_k, a)\, v^*(s_j) \right) \right] \\ &= L\underline{v}^*(s_k), \end{aligned}$$

thus proving that

$$\underline{v}^* = G\underline{v}^* \implies \underline{v}^* = L\underline{v}^*.$$

But we know that L has a unique fixed point $v^*_{\infty,\lambda}$. Thus,

$$\underline{v}^* = v^*_{\infty,\lambda}, \qquad \underbrace{\underline{v}_n \to v^*_{\infty,\lambda}}_{\|\underline{v}_n - v^*_{\infty,\lambda}\|_\infty \to 0 \text{ as } n \to \infty.} \text{ as } n \to \infty.$$

(2) Follows by using $sp(\underline{v}) \leq 2\|\underline{v}\|_\infty$.

(3) Part (3) is just a consequence of part (1) [definition of convergence].

(4a) Recall the policy ψ_ε returned by the algorithm:

$$\psi_\varepsilon(s_k) = \arg\max_a \left[r(s_k, a) + \lambda \left(\sum_{j<k} P(s_j | s_k, a)\, v'(s_j) + \sum_{j \geq k} P(s_j | s_k, a)\, v(s_j) \right) \right]$$

Given $\phi : \mathcal{S} \to \Delta(\mathcal{A})$, define $G_\phi : \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ as

$$G_\phi \underline{v}(s_k) = \sum_a \phi(a | s_k) \left[r(s_k, a) + \lambda \left(\sum_{j<k} P(s_j | s_k, a)\, G_\phi \underline{v}(s_j) + \sum_{j \geq k} P(s_j | s_k, a)\, v(s_j) \right) \right]$$

Exercise: (1) Show that $G_\phi : \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ is a contraction with $v^\phi_{\infty,\lambda}$ as fixed point.

(2) Show that $G\underline{v}_n = G_{\psi_\varepsilon} \underline{v}_n$ $\forall n \geq 0$.

It then follows that

$$\| v^{\psi_\varepsilon}_{\infty,\lambda} - v^*_{\infty,\lambda} \|_\infty \le \| v^{\psi_\varepsilon}_{\infty,\lambda} - G_{\psi_\varepsilon} \underline{v}_n \|_\infty + \| G_{\psi_\varepsilon} \underline{v}_n - v^*_{\infty,\lambda} \|_\infty$$

$$= \| G_{\psi_\varepsilon} v^{\psi_\varepsilon}_{\infty,\lambda} - G_{\psi_\varepsilon} \underline{v}_n \|_\infty + \| G \underline{v}_n - G \underline{v}^*_{\infty,\lambda} \|_\infty$$

$$\le \lambda \| v^{\psi_\varepsilon}_{\infty,\lambda} - \underline{v}_n \|_\infty + \lambda \| \underline{v}_n - \underline{v}^*_{\infty,\lambda} \|_\infty \to (a)$$

Now, we have

$$\| v_n - v^*_{\infty,\lambda} \|_\infty \le \| \underline{v}_{n+1} - \underline{v}_n \|_\infty + \| \underline{v}_{n+1} - v^*_{\infty,\lambda} \|_\infty$$

$$= \| \underline{v}_{n+1} - \underline{v}_n \|_\infty + \| G \underline{v}_n - G \underline{v}^*_{\infty,\lambda} \|_\infty$$

$$\le \| \underline{v}_{n+1} - \underline{v}_n \|_\infty + \lambda \| \underline{v}_n - v^*_{\infty,\lambda} \|_\infty$$

$$\Rightarrow \| \underline{v}_n - v^*_{\infty,\lambda} \|_\infty \le \frac{1}{1-\lambda} \| \underline{v}_{n+1} - \underline{v}_n \|_\infty . \to (1)$$

Along similar lines,

$$\| v^{\psi_\varepsilon}_{\infty,\lambda} - \underline{v}_n \|_\infty \le \| \underline{v}_{n+1} - \underline{v}_n \|_\infty + \| \underline{v}_{n+1} - v^{\psi_\varepsilon}_{\infty,\lambda} \|_\infty$$

$$= \| \underline{v}_{n+1} - \underline{v}_n \|_\infty + \| G_{\psi_\varepsilon} \underline{v}_n - G_{\psi_\varepsilon} v^{\psi_\varepsilon}_{\infty,\lambda} \|_\infty$$

$$\le \| \underline{v}_{n+1} - \underline{v}_n \|_\infty + \lambda \| \underline{v}_n - v^{\psi_\varepsilon}_{\infty,\lambda} \|_\infty$$

$$\Rightarrow \| v^{\psi_\varepsilon}_{\infty,\lambda} - \underline{v}_n \|_\infty \le \frac{1}{1-\lambda} \| \underline{v}_{n+1} - \underline{v}_n \|_\infty . \to (2)$$

Putting (1) and (2) back into (a), we get

$$\| v^{\psi_\varepsilon}_{\infty,\lambda} - v^*_{\infty,\lambda} \| \le \frac{2\lambda}{1-\lambda} \| \underline{v}_{n+1} - \underline{v}_n \|_\infty .$$

Upon stoppage, we have $\| \underline{v}_{n+1} - \underline{v}_n \|_\infty = \sigma < \frac{(1-\lambda)}{\lambda} \varepsilon$

$$\Rightarrow \| v^{\psi_\varepsilon}_{\infty,\lambda} - v^*_{\infty,\lambda} \| \le 2\varepsilon$$

(4b) Follows from eq. (1) ; upon stoppage, $\| \underline{v}_{n+1} - \underline{v}_n \| < \left(\frac{1-\lambda}{\lambda}\right) \varepsilon$.

Asynchronous Value Iteration

At time $n+1$, if we decide to update the value of state $s = s''$, then

$$v_{n+1}(s) = \begin{cases} \max_a \left[r(s,a) + \lambda \sum_{s' \in \mathcal{S}} P(s'|s,a)\, v_n(s') \right], & s = s'', \\ v_n(s), & s \neq s''. \end{cases}$$

(One state update per time epoch n)